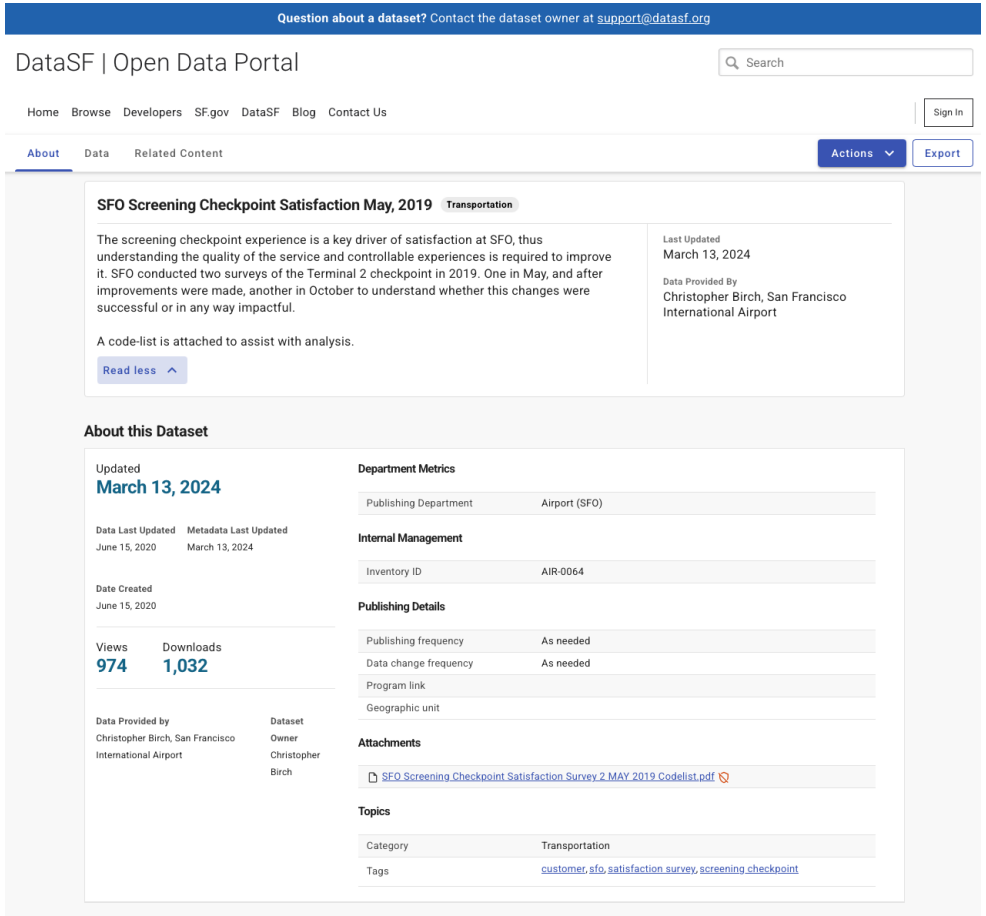
Public Portfolio Series: SFO Customer Experience

Choosing the right dataset



- Public data enabling sharable content
- Pre- and post- intervention structure
- Professionally-designed questionnaire
- Customer experience focus that aligns with my interests
- Relatively few downloads, so an opportunity for a distinctive analysis

Setting up for success

Background & Data Dictionary

Monday, February 2, 2026 7:39 AM

URL	https://catalog.data.gov/dataset/sfo-screening-checkpoint-satisfaction-may-2019
URL	https://catalog.data.gov/dataset/sfo-screening-checkpoint-satisfaction-october-2019
SF May URL	https://data.sfgov.org/Transportation/SFO-Screening-Checkpoint-Satisfaction-May-2019/jt6x-6hpy/about_data
SF Oct URL	https://data.sfgov.org/Transportation/SFO-Screening-Checkpoint-Satisfaction-October-2019/xyey-v962/about_data
Location of Stored Data	/Volumes/LaCie/SFO_Satisfaction

Field	Code/Description	Type	Valid Values
RESPNUM	Respondent Number – automatically generated during data entry		
CCGID	CCG ID assigned durng batching		
RUNID	Run on which survey was collected		
METH	Method of collection		1 - At Gate 2 - Mail-in
Q1	Did you go through security (e.g. security checkpoint) to enter this terminal today?		1 - Yes 2 - No
Q2	How would you rate your SFO security checkpoint experience on each of the following:		5 - Excellent 4 3 2 1 - Poor 6 - Don't Know 0 - Blank
Q2A	Courtesy of checkpoint entry staff		
Q2B	Courtesy of ID check podium staff		
Q2C	Courtesy and helpfulness of checkpoint screening staff		
Q2D	Efficiency of screening staff		
Q2E	Signage to checkpoint		
Q2F	Checkpoint organization		
Q2G	Waiting time in security line		
Q2H	Noise level		
Q2I	Music selection and volume		
Q2J	Space allotted for you to prepare to go through screening (where luggage is placed on conveyor, etc.)		
Q2K	Cleanliness of checkpoint area		
Q2L	Repacking area (once you've passed through security)		
Q3	Overall rating of security checkpoint		5 - Excellent 4 3 2 1 - Poor 6 - Don't Know 0 - Blank
Q3A_VERBATIM	Why is that		
Q4_1	Which of the following lines/areas did you use to enter the security checkpoint today?		1 - General / main security line 2 - PreCheck 3 - Premium 4 - CLEAR 5 - Other (specify) 6 - Don't Know 7 - Blank 8 - Disabled/special assistance (added by respondents)
Q5	How safe did you feel going through the security checkpoint at SFO today?		
Q6	What is the primary purpose of your trip?		1 - Business 2 - Personal 3 - Other (specify) 0 - Blank
Q7	About how many times have you flown out of SFO in the past 12 months?		1 - 1 2 - 2

00B-SFO_DATA_TRANSLATION.sas × 00A-SFO_MAIN_IMPORT.sas × *01A-SFO_INITIAL_VARIABLE_SCAN.sas ×

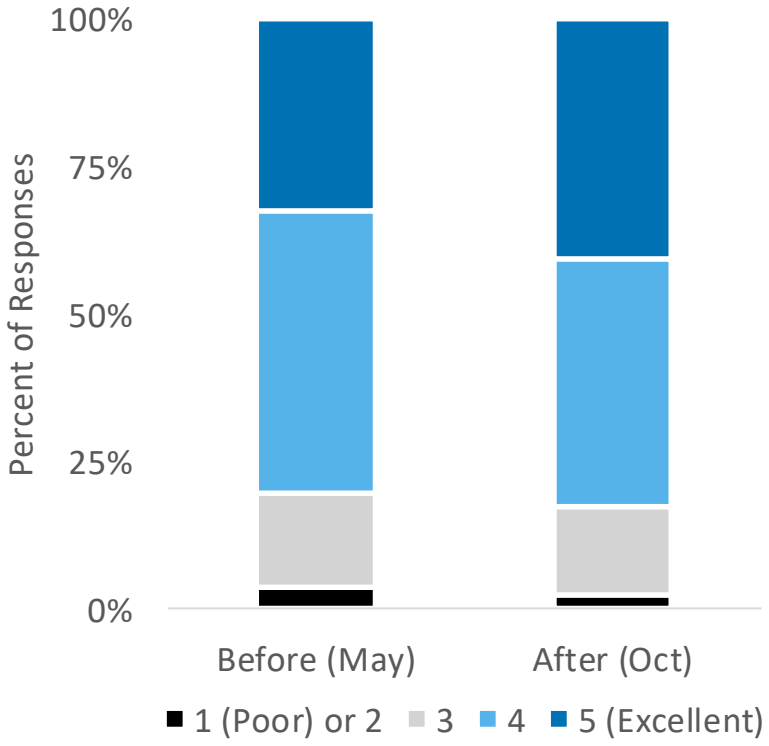
CODE LOG RESULTS OUTPUT DATA

```
/*Date:          17JAN2026*/
/*Analyst:       Beau Brouillette*/
/*Description:   The purpose of this script is to format the raw data from the import (00A-SFO_MAIN_IMPORT.sas) so that MAY and OCT responses
                 are combined into a single dataset and the encoding of variables is human-readable (rather than numeric codes).
                 Suffix _TX will be used to denote transformed/translated varaibles.
                 _TX versions of variables will be encoded based on detail from data dictionary (codelist PDF) that accompanies these datasets.
                 A review of the PDFs shows consistent encoding of variables between the MAY and OCT datasets.

                 The final portion of this script will complete some quality assurance to ensure that the expected number of records is found in the dataset
                 and that the data from the raw import variables is consistently translated into human-readable format and that translation matches
                 expectations for that variable.

                 The City and County of San Francisco has made some customer survey information publically available.
                 These surveys capture customer satisfaction survey responses before and after an update to the TSA
                 screening area at one of the San Francisco International Airport (SFO) terminals in 2019.

                 Details on the source of this data are provided below:
                     SFO May URL: https://data.sfgov.org/Transportation/SFO-Screening-Checkpoint-Satisfaction-May-2019/jt6x-6hpy/about_data
                     SFO Oct URL: https://data.sfgov.org/Transportation/SFO-Screening-Checkpoint-Satisfaction-October-2019/xyey-v962/about_data*/

proc sql;
   create table SFO_PRE_POST_SURVEY as
       select TIMEFRAME,   /*User-defined variable TIMEFRAME with levels PRE = May 2019 pre-intervention responses, POST = October 2019 post-intervention responses*,

           /*ID variables from input files*/
           RESPNUM,          /*Respondent Number – automatically generated during data entry*/
           CCGID,            /*CCG ID assigned durng batching*/
           RUNID,            /*Run on which survey was collected*/
           AREA,             /*[Not defined in codelist data dictionary. Retaining, but appears to be all 'D', so uninformative]

           /*Day and Date detail from raw import*/
           DAY,
           DATE format = date9., /*Need to specify DATE9 (ddMONyyyy) format because this did not come through in the import process*/

           /*METH: Method of collection*/
           METH,                 /*Method of collection*/
           /*Use case logic to encode description from data dictionary*/
           case    when METH = 1            then '1 - At Gate'
                   when METH = 2            then '2 - Mail-in'
                   else '0 - Other/Unk/Missing' end    as METH_TX,

           /*Q1: Did you go through security (e.g. security checkpoint) to enter this terminal today?*/
           Q1,
           case    when Q1 = 1              then '1 - Yes'
                   when Q1 = 2              then '2 - No'
                   else '0 - Other/Unk/Missing' end    as Q1_TX,

           /*Q2:How would you rate your SFO security checkpoint experience on each of the following:*/
               /*Data dictionary shows this encoding of Q2 and Q3 variables:
                   5 Excellent
                   4
                   3
                   2
                   1 Poor
                   6 Don't know
                   0 Blank*/
               /*As a first pass, I will create the TX version with 6/0/[null] combined into [null].
                   I will also populate descriptors for levels 2-4, though it looks like these were not present on the survey and are imputed for convenience*/
           /*Q2A: Courtesy of checkpoint entry staff*/
           Q2A,
           case    when Q2A = 5             then '5 - Excellent'
                   when Q2A = 4             then '4 - Satisfactory'
                   when Q2A = 3             then '3 - Neutral'
                   when Q2A = 2             then '2 - Unsatisfactory'
                   when Q2A = 1             then '1 - Poor'
                   when Q2A in (0,6)        then '0 - Other/Unk/Missing'
                   else '0 - Other/Unk/Missing' end as Q2A_TX,
           /*Q2B: Courtesy of ID check podium staff*/
           Q2B,
           case    when Q2B = 5             then '5 - Excellent'
                   when Q2B = 4             then '4 - Satisfactory'
                   when Q2B = 3             then '3 - Neutral'
                   when Q2B = 2             then '2 - Unsatisfactory'
                   when Q2B = 1             then '1 - Poor'
                   when Q2B in (0,6)        then '0 - Other/Unk/Missing'
                   else '0 - Other/Unk/Missing' end as Q2B_TX,
           /*Q2C: Courtesy and helpfulness of checkpoint screening staff*/
           Q2C,
           case    when Q2C = 5             then '5 - Excellent'
                   when Q2C = 4             then '4 - Satisfactory'
                   when Q2C = 3             then '3 - Neutral'
                   when Q2C = 2             then '2 - Unsatisfactory'
                   when Q2C = 1             then '1 - Poor'
                   when Q2C in (0,6)        then '0 - Other/Unk/Missing'
                   else '0 - Other/Unk/Missing' end as Q2C_TX,
           /*Q2D: Efficiency of screening staff*/
           Q2D,
           case    when Q2D = 5             then '5 - Excellent'
                   when Q2D = 4             then '4 - Satisfactory'
                   when Q2D = 3             then '3 - Neutral'
                   when Q2D = 2             then '2 - Unsatisfactory'
```

Satisfaction increased after upgrades

SFO CX

Question 3: "Overall rating of security checkpoint"



Mean score increased 4.08 → 4.21

Modest, but statistically significant

Correlated responses suggest meaningful lift in key areas

Similar improvement in 2 of the 4 biggest drivers of satisfaction

Rank	Wording	Mean Score (Before → After)
1 ★	"Efficiency of screening staff"	4.21 → 4.36+
2	"Checkpoint organization"	4.12 → 4.22
3	"Cleanliness of checkpoint area"	4.11 → 4.14
4 ★	"Courtesy and helpfulness of checkpoint screening staff"	4.19 → 4.34*

Recommendations for the future

Analysis of customer comments aligned with satisfaction ratings

Rank	Wording	Mean Score (Before → After)
1 ★	"Efficiency of screening staff"	4.21 → 4.36+
2	"Checkpoint organization"	4.12 → 4.22
3	"Cleanliness of checkpoint area"	4.11 → 4.14
4 ★	"Courtesy and helpfulness of checkpoint screening staff"	4.19 → 4.34*

Add more repacking space

Maintain gains that may have come from training

Increase cleaning of high-touch surfaces